15

Non-Ohmic hopping conduction and electronic switching

Driving a steady-state current of carriers hopping between inequivalent states generally alters sites' occupancies. This carrier redistribution grows as the temperature is lowered or the current is increased. Carriers' redistribution becomes significant when the hopping conduction becomes non-Ohmic.

This chapter begins by describing how the current-versus-voltage relationship is determined for steady-state hopping conduction among inequivalent states. As flow becomes arbitrarily weak (1) carrier redistribution becomes ignorable, (2) charge transport becomes Ohmic, and (3) hopping transport reduces to that of a resistor network. The remainder of the chapter focuses on the carrier redistribution that accompanies non-Ohmic hopping.

Current-driven carrier redistribution of charge carriers is explored through two examples. Section 15.2 addresses low-temperature hopping among inequivalent states of a disordered medium. Section 15.3 describes effects of carrier transfers between a small-polaron semiconductor and non-polaronic electrical contacts. Increasing the steady-state current augments the densities of slow-moving small polarons which accumulate near the semiconductor's interfaces with its contacts. A sufficiently strong current may drive the density of interfacial small polarons high enough to preclude their formation. This effect suggests a mechanism for reversibly switching much of the semiconductor into a relatively high-conductance state.

15.1 Formalism for steady-state hopping

The net electric current produced by carriers of charge q hopping between sites i and j is

$$I_{i,j} = q\left[f_i(1-f_j)R_{i,j} - f_j(1-f_i)R_{j,i}\right], \tag{15.1}$$

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where f_i and f_j are the probabilities of sites i and j being occupied and $R_{i,j}$ and $R_{j,i}$ are the rates with which a carrier jumps from site i to site j and from site j to site i, respectively. A site's occupation probability can be expressed in terms of a function which resembles the Fermi distribution function:

$$f_i \equiv \frac{1}{\exp[(E_i - \mu_i)/\kappa T] + 1}$$

$$= \frac{1}{2}\exp\left(-\frac{E_i - \mu_i}{2\kappa T}\right)\operatorname{sech}\left(\frac{E_i - \mu_i}{2\kappa T}\right) = \frac{p_i F_i}{1 + p_i F_i}, \qquad (15.2)$$

where μ_i, the local quasi-electro-chemical potential of site i replaces μ_c, the electro-chemical potential of an equilibrated system. Here E_i denotes the energy of a carrier at site i in the presence of an applied electric field and κT represents the thermal energy. Following the final equality of Eq. (15.2), f_i is expressed in terms of the local quasi-fugacity $F_i \equiv \exp(\mu_i/\kappa T)$ and the corresponding probability factor $p_i \equiv \exp(-E_i/\kappa T)$. Analogously,

$$1 - f_j \equiv \frac{1}{1 + \exp\left[-\left(E_j - \mu_j\right)/\kappa T\right]}$$

$$= \frac{1}{2}\exp\left(\frac{E_j - \mu_j}{2\kappa T}\right)\operatorname{sech}\left(\frac{E_j - \mu_j}{2\kappa T}\right) = \frac{1}{1 + p_j F_j}. \qquad (15.3)$$

With thermally equilibrated phonons, the rate for a carrier's phonon-assisted hop from site i to site j assumes the general form (Emin, 1974)

$$R_{i,j} = \exp\left[-\left(E_j - E_i\right)/2\kappa T\right] r_{i,j}\left(\left|E_j - E_i\right|\right) = \sqrt{\frac{p_j}{p_i}}\, r_{i,j}\left(\left|E_j - E_i\right|\right), \qquad (15.4)$$

where $r_{i,j}(|E_j - E_i|) = r_{j,i}(|E_i - E_j|)$. This rate and that for the reverse process manifestly satisfy the requirement of detailed balance:

$$R_{j,i}/R_{i,j} = \exp\left[\left(E_j - E_i\right)/\kappa T\right] = p_i/p_j. \qquad (15.5)$$

With Eqs. (15.2)–(15.4) incorporated into Eq. (15.1) it becomes evident that current flow between a pair of sites depends upon the differences of their quasi-electro-chemical potentials and quasi-fugacities, $\mu_i - \mu_j$ and $F_i - F_j$, respectively:

$$I_{i,j}=\frac{q}{2}r_{i,j}(|E_j-E_i|)\sec h\left(\frac{E_i-\mu_i}{2\kappa T}\right)\mathrm{sech}\left(\frac{E_j-\mu_j}{2\kappa T}\right)\sin\,\mathrm{h}\left(\frac{\mu_i-\mu_j}{2\kappa T}\right)$$
$$=qr_{i,j}(|E_j-E_i|)\frac{\sqrt{p_ip_j}}{(1+p_iF_i)(1+p_jF_j)}(F_i-F_j). \tag{15.6}$$

With steady-state flow, the carrier density at each site becomes time independent as the net current flowing to it vanishes:

$$\sum_j I_{i,j}=0, \tag{15.7}$$

for each site labeled by i. Equation (15.7) can be solved for the sets of μ_i or of F_i, where the values of these quantities at the boundaries of a sample are determined by the emf imposed across it. The quasi-electro-chemical potential and quasi-fugacity at each site depend on the site energies and their shifts with the applied electric field. With these values Eq. (15.6) can be used to obtain the current between any pair of sites as a function of the applied field. Of course, no current flows in equilibrium since the quasi-electro-chemical potential and the quasi-fugacity at every site then revert to their equilibrium values, μ_c and $F_c\equiv\exp(\mu_c/\kappa T)$, respectively.

When the applied electric field and the associated inter-site differences of the quasi-electro-chemical potentials are sufficiently small, $\mu_i-\mu_j\ll\kappa T$, the currents of Eq. (15.6) become simply proportional to these differences (Miller and Abrahams, 1960). The expression after the first equality in Eq. (15.6) then reduces to Ohm's law: $I_{i,j}=G_{i,j}(\mu_i-\mu_j)$, with the conductance

$$G_{i,j}\equiv\frac{q}{4\kappa T}r_{i,j}\left(\left|E_j^0-E_i^0\right|\right)\mathrm{sech}\left(\frac{E_i^0-\mu_c}{2\kappa T}\right)\sec\mathrm{h}\left(\frac{E_j^0-\mu_c}{2\kappa T}\right), \tag{15.8}$$

where the superscript 0 indicates local energies in the absence of the applied electric field. If the differences of the quasi-electro-chemical potentials for all significant hops of a system satisfy this condition, its hopping can be treated as a resistance network. However, this condition cannot be fulfilled if the temperature is too low or the current is too strong. The remainder of this chapter will discuss examples with such non-Ohmic behavior.

15.2 Low-temperature hopping in a disordered medium

Steady-state current flow drives a redistribution of sites occupied by hopping charge carriers. This effect becomes especially pronounced at low temperatures. To

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illustrate this effect and its implications for charge transport, consider just the current produced by a low density of charge carriers hopping between nearest neighbors along a linear chain of N sites.

With the presumption of a low carrier density, $p_iF_i \ll 1$, the expression for the current flowing between a pair of sites, Eq. (15.6), reduces to (Emin, 2006b)

$$I_{i,j} = qr_{i,j}\left(\left|E_j - E_i\right|\right)\sqrt{p_i p_j}\left(F_i - F_j\right). \tag{15.9}$$

Using this equation the condition for steady-state nearest-neighbor hopping at each interior chain site, $2 \leq i \leq N{-}1$, Eq. (15.7) becomes

$$\frac{F_{i-1} - F_i}{F_i - F_{i+1}} = \frac{r_{i,i+1}\left(\left|E_{i+1} - E_i\right|\right)}{r_{i-1,i}\left(\left|E_i - E_{i-1}\right|\right)}\sqrt{\frac{p_{i+1}}{p_{i-1}}} \equiv a_i. \tag{15.10}$$

The positive sign of the right-hand side of this equation for all values of i implies that the quasi-fugacity F_i changes monotonically along the chain. The quasi-fugacities at the ends of the chain, F_1 and F_N, are established by the boundary conditions for steady-state flow: the zero of potential energy and the imposed emf.

The quasi-fugacity at an intervening site s along the chain is determined by iterative solution of Eq. (15.10) to be (Emin, 2006b):

$$F_s = \frac{F_1 G_s + F_N\left(G_1 - G_s\right)}{G_1} \tag{15.11}$$

with $G_N \equiv 0$, where

$$G_s \equiv 1 + \sum_{i=s+1}^{N-1} \prod_{j=i}^{N-1} a_j \tag{15.12}$$

for $1 \leq s < N$. The steady-state current for nearest-neighbor hopping along the chain is found by inserting Eqs. (15.11) and (15.12) into Eq. (15.9):

$$I = I_{N-1,N} = \frac{qr_{N-1,N}\left(\left|E_N - E_{N-1}\right|\right)\sqrt{p_N p_{N-1}}}{1 + \sum_{i=2}^{N-1} \prod_{j=i}^{N-1} a_j}\left(F_1 - F_N\right). \tag{15.13}$$

The quasi-fugacity difference $F_1 - F_N$ drives the current. There are $N{-}1$ contributions to the denominator of Eq. (15.13). In the Ohmic limit each contribution is proportional to the corresponding resistance of an $(N{-}1)$-link chain of resistors.

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Beyond the Ohmic limit, the electric-field dependences of factors on the right-hand side of Eq. (15.13) other than $F_1 - F_N$ are no longer ignorable.

Attention is now focused on low-temperature hopping along the one-dimensional chain. In the low-temperature limit the net energy difference for a phonon-assisted hop upward in energy becomes the jump rate's activation energy. Meanwhile the rate for a hop downward in energy becomes temperature independent since it occurs via the spontaneous emission of phonons (Miller and Abrahams, 1960; Emin, 1974; 1975a; Bergeron and Emin, 1977). These two features emerge from the low-temperature limit of the general expression for a phonon-assisted jump rate Eq. (15.4) c.f. Eq. (11.20).

$$\lim_{T\to 0} r_{i,j}\left(\left|E_j - E_i\right|\right) \propto \exp\left(-\left|E_j - E_i\right|/2\kappa T\right). \tag{15.14}$$

The remaining factors in $r_{i,j}(|E_j - E_i|)$ depend on the specifics of the electron–phonon interaction. Indeed, the low-frequency ac conductivity of adiabatic small-polaron hopping at low temperatures, dominated by hops downward in energy, is determined by these dependences (Emin, 1992). Nonetheless, these dependences have been disregarded when addressing the low-temperature limit of the dc conductivity which is dominated by hops upward in energy (Ambegaokar *et al.*, 1971). With this approximation, $r_{i,j}(|E_j - E_i|) = r_0 \exp(-|E_j - E_i|/2\kappa T)$ with r_0 being just a constant. Upon adopting this simplification the site coefficients defined in Eq. (15.10) become

$$a_i = \frac{e^{-\left|E_{i+1}-E_i\right|/2\kappa T}}{e^{-\left|E_i-E_{i-1}\right|/2\kappa T}} \frac{e^{-E_{i+1}/2\kappa T}}{e^{-E_{i-1}/2\kappa T}}. \tag{15.15}$$

Evaluating Eq. (15.15) for the four possible variations of the energy with site index gives: (1) $a_i = \exp[-(E_{i+1} - E_i)/\kappa T]$ for $E_{i-1} < E_i < E_{i+1}$, (2) $a_i = 1$ for $E_{i-1} < E_i > E_{i+1}$, (3) $a_i = \exp[-(E_i - E_{i-1})/\kappa T]$ for $E_{i-1} > E_i > E_{i+1}$, and (4) $a_i = \exp[-(E_{i+1} - E_{i-1})/\kappa T]$ for $E_{i-1} > E_i < E_{i+1}$.

Lowering the temperature or increasing the steady-state current produced by carriers' hopping among energetically inequivalent sites alters their occupancies. Concomitantly the relationship between the current and its electromotive force becomes increasingly nonlinear. The simplest example of hopping along a linear chain is used to illustrate these features.

Figure 15.1 depicts a succession of four hops which transport a carrier over an energy barrier. The energies of the first and fifth sites of the five-site chain are chosen to equal one another in the absence of the applied field. Since the energy of a carrier includes the potential energy of the imposed electric field, the difference between the energy of a carrier at site 1 and at site 5 is just the product of the applied potential V and the carrier charge q: $E_1 - E_5 = qV$.

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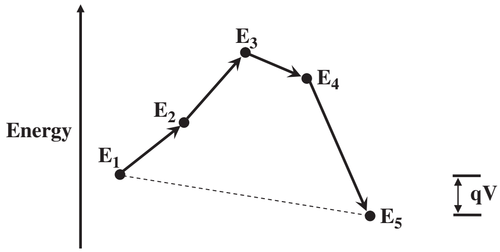

Fig. 15.1 This illustration depicts the energies for the four nearest-neighbor hops via which the emf qV drives a carrier over an energy barrier on a five-atom linear chain.

Imposing the electromotive force establishes the boundary conditions on the quasi-fugacities at sites 1 and 5: $F_1 = 1$ and $F_5 = \exp(-qV/\kappa T)$. The quasi-fugacities at sites 2, 3, and 4 are found from Eqs. (15.11) and (15.12). Then the model of Eq. (15.15) is utilized to express the a_i in terms of the E_i. The results are:

$$\begin{aligned} F_2 &= \frac{F_5(a_2a_3a_4) + F_1(a_3a_4 + a_4 + 1)}{(a_2a_3a_4 + a_3a_4 + a_4 + 1)} \\ &= \frac{e^{-qV/\kappa T}e^{(E_2-E_4)/\kappa T} + 2e^{(E_3-E_4)/\kappa T} + 1}{e^{(E_2-E_4)/\kappa T} + 2e^{(E_3-E_4)/\kappa T} + 1}, \end{aligned} \tag{15.16}$$

$$\begin{aligned} F_3 &= \frac{F_5(a_2a_3a_4 + a_3a_4) + F_1(a_4 + 1)}{(a_2a_3a_4 + a_3a_4 + a_4 + 1)} \\ &= \frac{e^{-qV/\kappa \mathrm{T}}\left[e^{(E_2-E_4)/\kappa T} + e^{(E_3-E_4)/\kappa T}\right] + e^{(E_3-E_4)/\kappa T} + 1}{e^{(E_2-E_4)/\kappa T} + 2e^{(E_3-E_4)/\kappa T} + 1}, \end{aligned} \tag{15.17}$$

and

$$\begin{aligned} F_4 &= \frac{F_5(a_2a_3a_4 + a_3a_4 + a_4) + F_1}{(a_2a_3a_4 + a_3a_4 + a_4 + 1)} \\ &= \frac{e^{-qV/\kappa T}\left[e^{(E_2-E_4)/\kappa T} + 2e^{(E_3-E_4)/\kappa T}\right] + 1}{e^{(E_2-E_4)/\kappa T} + 2e^{(E_3-E_4)/\kappa T} + 1}. \end{aligned} \tag{15.18}$$

Figure 15.2 depicts the quasi-electro-chemical potential for three distinct limiting regimes. (1) When $\kappa T \gg E_3 - E_4, E_4 - E_2, qV$, the quasi-electro-chemical potential $\mu_i \equiv \kappa T \ln(F_i)$ varies almost uniformly along the chain. Then, as depicted by the dotted line in Fig. 15.2, $\mu_1 = 0$, $\mu_2 = -qV/4$, $\mu_3 = -qV/2$, $\mu_4 = -3qV/4$, $\mu_5 = -qV$. (2) As the temperature is lowered the thermal energy falls below the electronic energy

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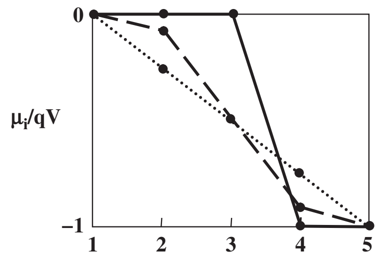

Fig. 15.2 The quasi-electro-chemical potentials μ_i in units of the driving emf qV are plotted at the five sites (large dots) involved in flow across the energy barrier depicted in Fig. 15.1. Three situations are described. (1) The values at the five sites are linked by the dotted line when the thermal energy κT greatly exceeds both the barrier height and the applied emf. (2) The values at the five sites are linked by the dashed line when κT is much less than the barrier height but much greater than qV. (3) The five sites are linked by the solid line in the low-temperature limit in which κT is much less than both the barrier height and qV.

differences, $qV \ll \kappa T \ll E_4 - E_2, E_3 - E_4$. Then, as shown by the dashed curve of Fig. 15.2, the variation of the quasi-electro-chemical potential occurs primarily in the vicinity of the barrier which limits carriers' flow. In particular, the dashed line in Fig. 15.2 shows the quasi-electro-chemical potentials for the energy barrier depicted in Fig. 15.1. The corresponding values are: $\mu_1 = 0$, $\mu_2 \approx 0$, $\mu_3 \approx -qV/2$, $\mu_4 \approx -qV$, $\mu_5 = -qV$. (3) Finally, as the temperature is lowered toward absolute zero $\kappa T \ll qV \ll E_4 - E_2, E_3 - E_4$. Then, as shown by the solid line of Fig. 15.2, the variation of the quasi-electro-chemical potential occurs primarily over the downstream portion of the barrier which limits carriers' flow. In particular, the solid line in Fig. 15.2 shows the low-temperature limits of the quasi-electro-chemical potentials for the energy barrier depicted in Fig. 15.1. The corresponding values are: $\mu_1 = 0$, $\mu_2 \approx 0$, $\mu_3 \approx -\kappa T \ln(2)$, $\mu_4 \approx -qV$, $\mu_5 = -qV$.

Steady-state current flow alters sites' occupation probabilities. The occupation probability of a site f_i relative to its equilibrium value f_i^0 is

$$f_i/f_i^0 = \exp\{[qV(i-1)/4 + \mu_i]/\kappa T\}. \tag{15.19}$$

Figure 15.3 displays $\ln(f_i/f_i^0)/(qV/\kappa T)$ plotted against the site index i for the three limits depicted in Fig. 15.2. As the temperature is lowered carriers participating in the flow illustrated in Fig. 15.1 increasingly accumulate on the upstream side of the energy barrier and are depleted on its downstream side.

The steady-state current through the energetic landscape depicted in Fig. 15.1 is obtained from Eq. (15.13) after writing the quasi-fugacities F_1 and F_5 in terms of the electromotive force and using the model of Eq. (15.15) to express the a_i coefficients in terms of the field-dependent carrier energies, the E_i.

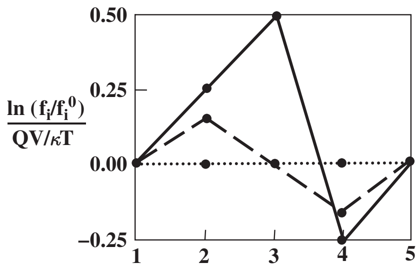

Fig. 15.3 The logarithm of the ratio of a site's occupation probability under steady-state flow f_i to that in equilibrium f_i^0 is plotted in units of the ratio of the emf to the thermal energy $qV/\kappa T$ for the five sites comprising the energy barrier of Fig. 15.1. As the temperature is lowered carriers increasingly accumulate in front of the barrier while being depleted behind it.

$$I = \frac{qr_0}{e^{E_2/\kappa T} + 2e^{E_3/\kappa T} + e^{E_4/\kappa T}} \left(1 - e^{-qV/\kappa T}\right)$$
$$\rightarrow qr_0 e^{-E_3^0/\kappa T} \sinh(qV/2\kappa T). \tag{15.20}$$

Here $E_3{}^0$ is the energy of the carrier at site 3 in the absence of the applied electric field and r_0 is the jump-rate constant introduced above Eq. (15.15). The expression after the arrow in Eq. (15.20) indicates that the current's low-temperature limit only depends on the energy-landscape's highest field-free energy $E_3{}^0$.

This current becomes non-Ohmic in the low-temperature limit $T \rightarrow 0$ while keeping V finite. Nonetheless, the low-temperature limit of a disordered medium's hopping conductivity was addressed under the presumption that it remains Ohmic, $I \propto V$ (Ambegaokar *et al.*, 1971). In particular, hopping transport was assumed to be describable by a linear-response resistor network with conductances given by Eq. (15.8). Such a limit corresponds to taking $T \rightarrow 0$ while also requiring that $\kappa T \gg qV$.

The one-dimensional example of this section illustrates how steady-state flow of hopping charge carriers in a disordered medium alters the probabilities of sites being occupied. This effect increases as (1) the temperature is lowered, (2) the current is raised and (3) the disorder is enhanced. Increasing the dimensionality quantitatively reduces the effects of disorder by offering more ways for carriers to avoid negotiating especially difficult jumps.

15.3 Interfacial small-polaron accretion: a threshold switching mechanism

Driving a steady-state current of small polarons can enhance their densities near its semiconductor's interfaces with high-mobility contacts. Two effects specific to small-polarons foster their accumulation near these interfaces. First, the mobilities

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of small polarons are generally much lower than those of carriers moving through conventional leads. Second, conversion between the semiconductor's small polarons and the leads' conventional carriers is slowed by the need to move the atoms associated with small-polaron formation. As the current is increased small-polarons' interfacial densities rise toward a maximum beyond which they can no longer form. Reaching this maximum is posited to trigger an avalanche instability in which all but a residual region of the semiconductor reversibly switches into a state of higher conductance.

A narrow-band model is utilized to illustrate the redistribution of carriers that accompanies their current-driven passages between high-mobility leads and a low-mobility small-polaron semiconductor (Emin, 2006b). As illustrated in Fig. 15.4 a linear chain of N sites is divided into a central portion of n small-polaron sites and two m-site leads, $N = n + 2m$. Disorder-related energy variations among each of the two classes are incidental and are therefore ignored. The energy at a small-polaron site labeled by i is then $-E_b - qV(i-1)/(N-1)$, where E_b is the small-polaron binding energy and $V/(N-1)$ is the drop of the applied potential between adjacent sites. Analogously, the energy at a lead site is $-E_f - qV(i-1)/(N-1)$, where E_f is a free-carrier's energy in the absence of the applied electric field.

Carriers are presumed to move only between adjacent sites on the chain. The rates for these inter-site transitions are written in the general form displayed in Eq. (15.4). However transfer processes involving the semiconductor and its leads are characterized by distinct rate factors $r_{i,i\pm1}(|E_{i\pm1}-E_i|)$. For phonon-assisted hops of a carrier between small-polaron sites $r_{i,i\pm1}(|E_{i\pm1}-E_i|) \equiv R_s$. For a carrier transferring between neighboring lead sites $r_{i,i\pm1}(|E_{i\pm1}-E_i|) \equiv R_l$. For a carrier moving from a small-polaron site to a neighboring lead site $r_{i,i\pm1}(|E_{i\pm1}-E_i|) \equiv R_{s,l}$. Similarly, $r_{i,i\pm1}(|E_{i\pm1}-E_i|) \equiv R_{l,s}$ for a carrier moving from a lead site to an adjacent small-polaron site.

The steady-state current I and the concomitant shifts of sites' occupancies f_i/f_i^0 are functions of these rate factors and the driving potential V. The procedure to establish these dependences becomes especially simple if the site occupancies are presumed small. Then I and f_i/f_i^0 are functions of every site's steady-state coefficient a_i and V with Eq. (15.10) expressing each a_i in terms of the rate factors and V.

The values of a_i are obtained from Eq. (15.10) for the model of Fig. 15.4. For the left-side lead, $2 \le i \le m-1$, $a_i = \exp[qV/(N-1)\kappa T] \equiv A$. For the two sites bridging the

Fig. 15.4 A line of N sites is divided into two sets of m sites (solid dots) which represent leads to a semiconductor of n sites (open dots) upon which small polarons can form.

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semiconductor's left interface with a lead, $a_m = A(R_{l,s}/R_l)\exp(-\Delta/2\kappa T)$ and $a_{m+1} = A(R_s/R_{l,s})\exp(-\Delta/2\kappa T)$, where $\Delta \equiv E_f - E_b$. For the semiconductor's interior sites, $m+2 \le i \le m+n-1$, $a_i = A$. For the two sites bridging the semiconductor's right interface with a lead, $a_{m+n} = A(R_{s,l}/R_s)\exp(\Delta/2\kappa T)$ and $a_{m+n+1} = A(R_l/R_{s,l})\exp(\Delta/2\kappa T)$ with $R_{s,l} = R_{l,s}$. For the right-side lead, $m+n+2 \le i \le N-1$, $a_i = A$. Thus the a_i-coefficients only depart from A at interfacial sites.

These modifications of interfacial sites' a_i-coefficients cause the probability of a site s being occupied, f_s, to depart from its equilibrium value f_s^0 under steady-state flow. The deviation of f_s/f_s^0 from unity results from a competition between the electric-field-dependences of the site's energy and its quasi-fugacity F_s (Emin, 2006b):

$$\frac{f_s}{f_s^0} = \exp\left[\frac{qV(s-1)/(N-1)}{\kappa T}\right]F_s = A^{s-1}F_s. \tag{15.21}$$

With the boundary conditions corresponding to imposing the emf qV, $F_1 = 1$ and $F_N = \exp(-qV/\kappa T) = A^{1-N}$, the quasi-fugacity at site s given by Eq. (15.11) becomes:

$$F_s = \frac{G_s + A^{1-N}(G_1 - G_s)}{G_1}. \tag{15.22}$$

Combining Eq. (15.22) with Eq. (15.21) yields

$$\frac{f_s}{f_s^0} = \frac{A^{s-1}G_s + A^{s-N}(G_1 - G_s)}{G_1}, \tag{15.23}$$

where G_s is obtained by inserting the coefficients determined in the preceding paragraph into Eq. (15.12).

This procedure, analogous to the calculation of Sec. 15.2, has been performed for this section's model (Emin, 2006b). Once the leads are long enough so that their lengths do not affect the accretion and depletion of small polarons within the semiconductor:

$$\begin{aligned}\frac{f_u}{f_u^0} &= A^{u-n-2} + \left(\frac{R_l}{R_s}e^{\Delta/\kappa T}\right)\left[1 - A^{u-n-1} - \delta_{u,1}\left(1 - A^{-1}\right)\right] \\ &\quad + \left(\frac{R_l}{R_{s,l}}e^{\Delta/2\kappa T}\right)\left(1 - A^{-1}\right)\left(A^{u-n-1} + \delta_{u,1}\right),\end{aligned} \tag{15.24}$$

where the n semiconductor sites are now labeled by u, $n \ge u \ge 1$. Small polarons' site occupations revert to their equilibrium values, $f_u \to f_u^0$, in the limits that (1) the

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applied electric field vanishes, $A \rightarrow 1$, or (2) interfacial disruptions of flow vanish, $R_s = R_l = R_{s,l}$ with $\Delta = 0$.

Small-polarons' accumulation within the semiconductor are driven by their relatively slow transfers (1) through it, $R_s \ll R_l$, and (2) with its leads, $R_{s,l} \ll R_l$. As seen from Eq. (15.24) these impediments to flow act like the erecting of barriers produced when the semiconductor's energy levels rise above those of the leads to give $\Delta > 0$.

The contribution to the increase of the small-polaron density which is proportional to R_l/R_s in Eq. (15.24) rises from its value at the incident interface, site 1, to a peak at site 2 and then falls as the exiting interface is approached. By contrast, the contribution to the increase of the small-polaron density which is proportional to $R_l/R_{s,l}$ in Eq. (15.24) falls from its value at the incident interface to a minimum at site 2 and then rises as the exiting interface is approached. This interfacial-transfer contribution gains in prominence at high fields due to its proportionality to the factor $(1-A^{-1}) \equiv 1-\exp[-qV/(N-1)\kappa T]$. Indeed, this contribution will ultimately dominate at high fields if $R_{s,l}\exp(\Delta/2\kappa T) \ll R_s$. Then, as illustrated in Fig. 15.5, increasing the emf can drive large accumulations of small polarons near the semiconductor's interfaces. The dashed curves of Fig. 15.5 show that this accretion of small polarons persists as the semiconductor's thickness is decreased. All told, high electric fields can drive large enhancements of a semiconductors'

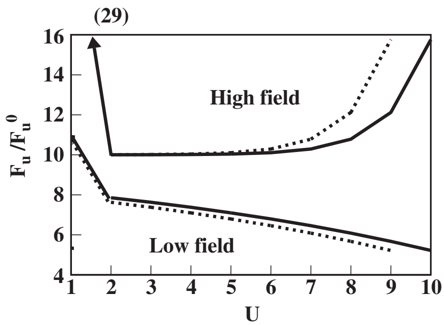

Fig. 15.5 The probability of site u on an n-site semiconductor chain being occupied during steady-state flow f_u relative to its equilibrium value f_u^0 is plotted against the site index u for carriers that enter the chain at site 1 and exit it from site n, $1 \le u \le n$. The two solid curves display f_u/f_u^0 for currents driven through a chain of $n = 10$ sites by high and low electric fields. These high- and low-field cases correspond to having potential differences between adjacent sites of κT and $\kappa T/10$, $\ln A = 1$ and 0.1 in Eq. (15.24), respectively. The two dashed curves are equivalent plots with the chain length reduced to $n = 9$. Carriers accumulate near the exiting interface when interfacial transfers between the semiconductor and the leads to it are impeded; here $(R_l/R_{s,l})\exp(\Delta/2\kappa T) = 40$ and $(R_l/R_s)\exp(\Delta/\kappa T) = 10$, where these physical parameters are defined in the text.

small-polaron density near their interfaces with high-mobility leads when charge transfers between them are severely restricted.

The rate associated with converting a carrier in a high-mobility lead into a small polaron in the semiconductor $R_{l,s}$ is reduced by the self-trapping of its carrier. As described in Section 4.3 and depicted in Fig. 4.3, a carrier introduced to a three-dimensional covalent material generally encounters a barrier to its self-trapping. Similarly, the rate associated with converting a small polaron in the semiconductor into a quasi-free carrier in a lead $R_{s,l}$ requires liberating the self-trapped carrier from the potential well which binds it. Such freeing of self-trapped carriers is slowed by the need to move the associated atoms. Thus conversions between quasi-free carriers in leads and small polarons in semiconductors are generally very slow.

Small-polaron semiconductors generally possess much larger densities of their low-mobility carriers than those of conventional semiconductors with their low densities of high-mobility carriers. Because of the slow conversion rates between a small-polaron semiconductors' low-mobility self-trapped carriers and the relatively high mobilities of leads' conventional carriers, strongly driven currents elevate small-polaron densities even higher.

However there is an upper limit to the permissible small-polaron density. In particular, as small-polarons' density is increased their carrier-induced deformations tend to interfere destructively with one another to limit the density for which stable small-polarons' formation is possible. The details of this interference depend on the specifics of the electron–phonon interactions, the phonons with which carriers interact and the material's structure (Emin, 1980b).

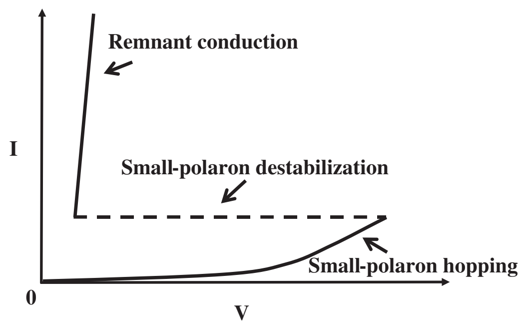

Fig. 15.6 An idealized I-versus-V curve for a semiconductor which manifests threshold switching shows a jump from low conductance (lower solid curve) to high conductance (upper solid curve) when flow exceeds a threshold. Reversion to low conductance occurs when high-conductance flow is sufficiently weakened. The model advanced here attributes the low-conductivity transport to the low-mobility hopping of a high density of small polarons. However, small polarons become destabilized when strong flow drives their density too high. The semiconductor then manifests the high conductance of the remnant.

https://doi.org/10.1017/CBO9781139023436.019 Published online by Cambridge University Press

Once the field is strong enough to drive an interfacial region's small-polaron density beyond its upper limit the region's carriers will no longer be self-trapped. Rather, the region's carriers will possess the greatly augmented mobility associated with quasi-free carriers. Thus the semiconductor's interfacial region will be converted into a free-carrier region. The net effect of this transformation just reduces the length of the semiconductor in which small polarons might form. However, Fig. 15.5 shows that simple shortening of the semiconductor shifts, but does not relieve, the field-driven elevation of the small-polaron density. This observation implies that a non-Ohmic strongly driven current will ultimately trigger an avalanche in which the semiconductor's carriers no longer form small polarons. As depicted in Fig. 15.6 this small-polaron destabilization provides a mechanism for threshold switching (Emin, 2006b).

Such switching phenomena are reported in many low-mobility solids in which small-polaron hopping is suspected, such as amorphous boron, transition-metal oxide glasses and chalcogenide glasses (Adler, 1971). For example, chalcogenide glasses manifest the high-temperature thermally activated mobilities and Hall-effect sign anomalies characteristic of small-polaron hopping (Emin *et al*., 1972; Seager *et al*., 1973; Seager and Quinn, 1975; Baily and Emin, 2006a; Baily *et al*., 2006b).

The flow-induced destabilization of small polarons is put forth as a general mechanism for threshold switching in small-polaron semiconductors. However, this mechanism by itself is insufficient to explain the many intricacies observed for threshold switching of real non-crystalline films. For example, the relatively high conductance of the switched state of a chalcogenide glass is generally independent of the nominal length of its conducting path (Adler *et al*., 1978; Fritzsche, 2006). In other words, the remnant conductance is not a bulk property of the switched glass.